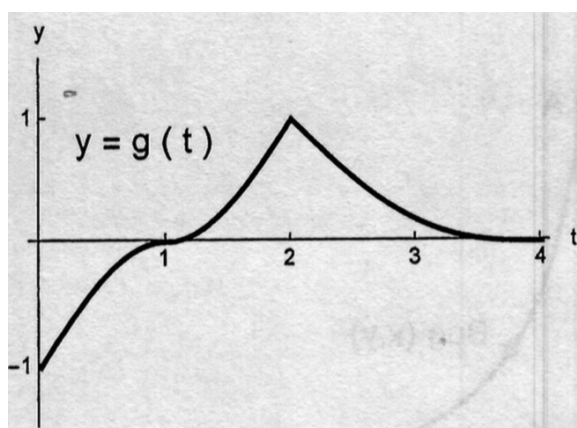


**Problem 7**

**Problem 7**

**Problem 1** The graph of  $g$ , a continuous function  $[0, 4]$ , is shown in the figure. Let  $A(x) = \int_0^x g(t) dt$  for  $0 \leq x \leq 4$ .



- (a) Circle the correct statement about  $A(1)$ .
- (i)  $A(1) = 0$ ;
  - (ii)  $A(1) < 0$ ;
  - (iii)  $A(1) > 0$ ;
  - (iv) none of the previous answers.
- (b) Circle the correct statement about  $A(1.5)$ .
- (i)  $A(1.5) = 0$ ;
  - (ii)  $A(1.5) < 0$ ;
  - (iii)  $A(1.5) > 0$ ;
  - (iv) none of the previous answers.
- (c) Circle the correct statement about  $A'(1.5)$ .

Problem 7

- (i)  $A'(1.5) = 0$ ;
- (ii)  $A'(1.5) < 0$ ;
- (iii)  $A'(1.5) > 0$ ;
- (iv) *none of the previous answers.*

(d) Circle the correct expression for  $\int_1^4 (g(t) + 1) dt$ .

- (i)  $A(4) + 1$ ;
- (ii)  $A(4) - A(1)$ ;
- (iii)  $A(4) - A(1) + 1$ ;
- (iv)  $A(4) + 3$ ;
- (v)  $A(4) - A(1) + 3$ ;
- (vi) *none of the previous answers.*

(e) Circle the correct statement about  $A(0)$ .

- (i)  $A(0) = 0$ ;
- (ii)  $A(0) = 1$ ;
- (iii)  $0 < A(0) < 1$ ;
- (iv)  $A(0) > 1$ ;
- (v)  $A(0) < 0$ ;
- (vi) *none of the previous answers.*

(f) Circle the interval (or intervals) where the function  $A$  is DECREASING.

- (i)  $(0, 1)$ ;
- (ii)  $(1, 2)$ ;
- (iii)  $(2, 4)$ ;
- (iv) *none of the previous answers.*

(g) Circle the value (or values) where the function  $A$  attains its MAXIMUM.

- (i)  $x = 0$ ;
- (ii)  $x = 1$ ;

Problem 7

- (iii)  $x = 2$ ;
  - (iv)  $x = 3$ ;
  - (v)  $x = 4$ ;
  - (vi) *none of the previous answers.*
- (h) *Circle the value (or values) where the function  $A$  attains its MINIMUM.*
- (i)  $x = 0$ ;
  - (ii)  $x = 1$ ;
  - (iii)  $x = 2$ ;
  - (iv)  $x = 3$ ;
  - (v)  $x = 4$ ;
  - (vi) *none of the previous answers.*
- (i) *Circle the interval (or intervals) where the function  $A$  is CONCAVE DOWN.*
- (i)  $(0, 1)$ ;
  - (ii)  $(1, 2)$ ;
  - (iii)  $(2, 4)$ ;
  - (iv) *none of the previous answers.*
- (j) *Circle the value (or values) of  $x$  where the function  $A$  has an inflection point.*
- (i)  $x = 1$ ;
  - (ii)  $x = 2$ ;
  - (iii)  $x = 3$ ;
  - (iv) *none of the previous answers.*
- (k) *Sketch the graph of  $A$ , based on items (e)-(j)*